

Algorithms and Complexity

- What is an algorithm
- Input, output, correctness
- Why efficiency matters

Asymptotic Notation

- Big-O, Omega, Theta
- Drop constants and lower terms
- Worst / average / best case

Arrays and Linked Lists

- Random vs sequential access
- Cost of insert and delete
- Memory locality

Stacks and Queues

- LIFO and FIFO
- Bracket matching, postfix
- Towards breadth-first search

Sorting 1 - Basics

- Selection, insertion, bubble
- What stable sorting means
- The meaning of $O(n^2)$

Sorting 2 - Divide and Conquer

- Merge sort
- Quicksort and the pivot
- Avoiding the worst case

Searching

- Linear search
- Preconditions of binary search
- Toward search trees

Hashing

- Properties of a hash function
- Collisions and chaining
- Load factor and rehashing

Graph Representation

- Adjacency matrix
- Adjacency list
- Dense vs sparse graphs

Graph Traversal

- Depth-first search
- Breadth-first search
- Visited marks and cycles

Shortest Paths

- Dijkstra
- Bellman-Ford
- When edges are negative

Dynamic Programming

- Optimal substructure
- Overlapping subproblems
- Memoization vs tabulation